Your buildings already pay for bad power. A.D.A. makes it visible and handles the routine.

A small controller and dashboard that knows when it's NEPA, gen or inverter, logs every generator hour, runs the borehole pump on tank level, and protects equipment from voltage spikes.

Made for Nigerian power

NEPA, generator and inverter changeovers are the normal case, not an exception.

Keeps working offline

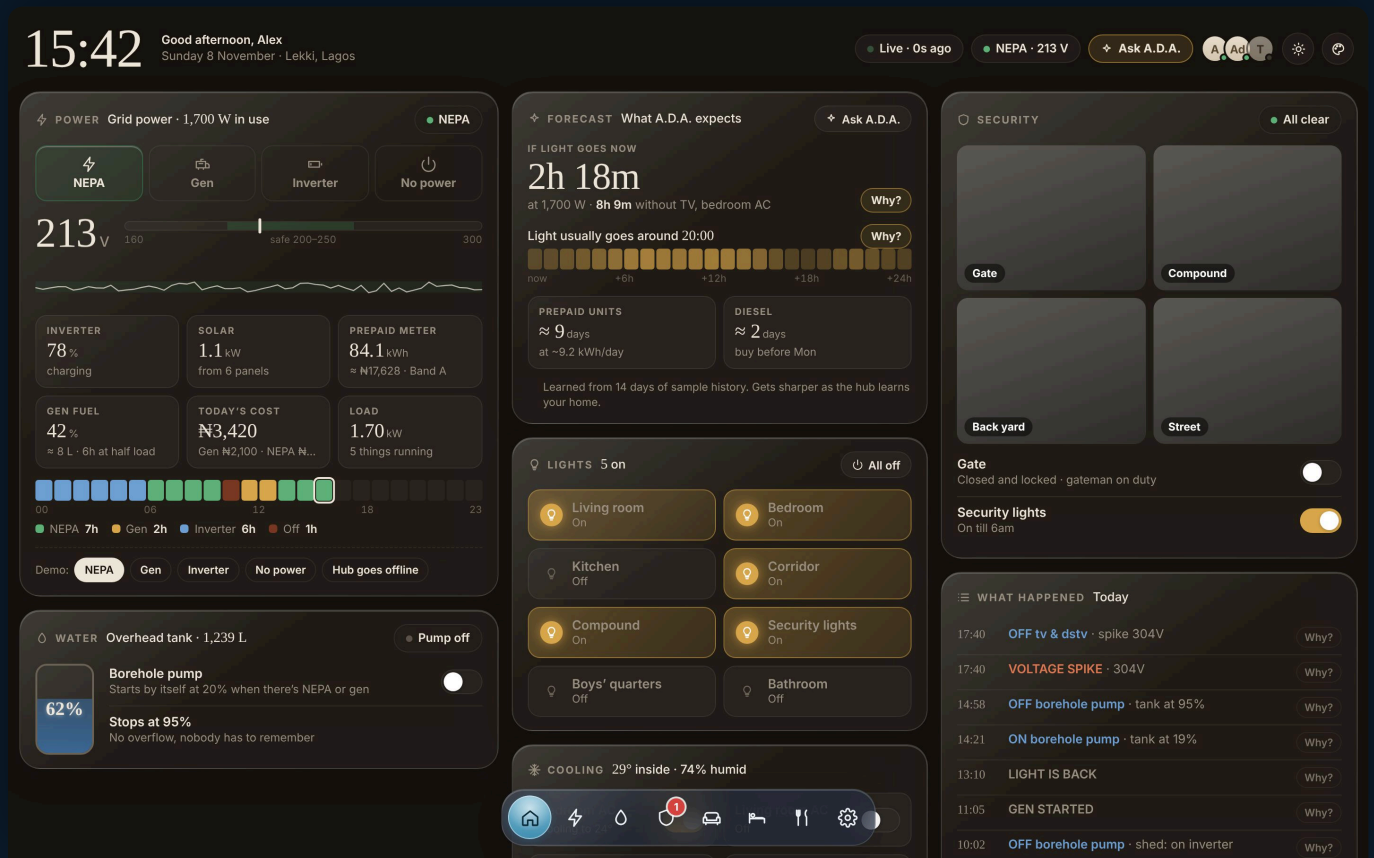
Automation runs on site. Internet is a bonus, not a requirement.

One view for every property

Status, forecasts and a plain log of what happened and why.

Prepared for **Max-Migold** · Pilot proposal

November 2026



THE PROBLEM

Power and water eat a facility manager's day

Across a portfolio, the same few issues repeat every week. They cost diesel, water, equipment and staff time, and most of them are hard to see until something goes wrong.

Diesel nobody can account for

Generator hours are tracked on paper, if at all. When the diesel bill comes, there's no reliable record of how long each gen ran or why, so reconciliation with vendors and residents turns into an argument.

Pumps left running, tanks run dry

Borehole pumps are switched by hand. Someone forgets, the tank overflows and wastes water and power. Or nobody switches it on before NEPA goes, and residents wake up to dry taps.

Batteries drained by heavy loads

When a building drops to inverter, ACs, pumping machines, irons and kettles keep running. The battery is flat by midnight and wears out years early. The installer gets blamed.

Equipment killed by voltage spikes

Grid and generator voltage swing well outside the safe range. TVs, fridges, CCTV and access control equipment fail long before their time, and nobody knows when it happened.

WHAT A.D.A. IS

Three parts that work together

The controller

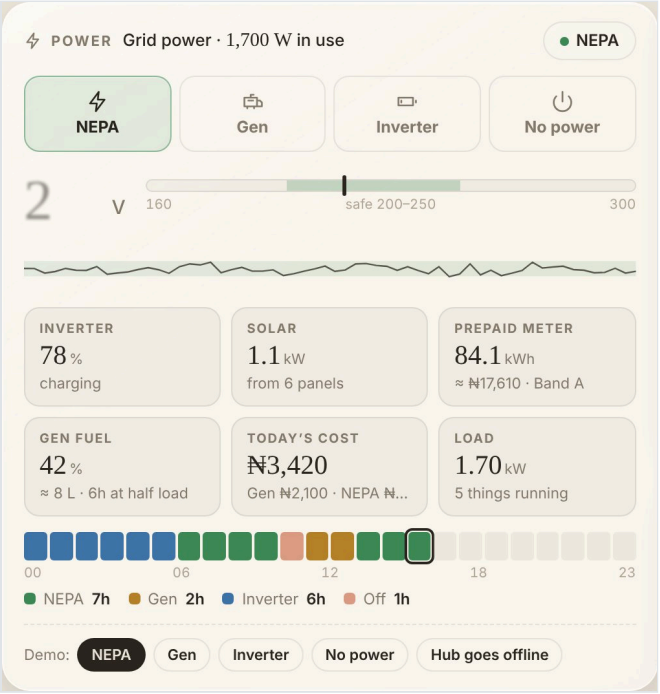
A small ESP32-based unit fitted near the distribution board. It reports the power source and switches the pump and selected heavy loads through relays. If it loses contact with the hub, relays drop to safe states.

The hub

Runs the automation rules and keeps the event log on site, so everything keeps working when the internet doesn't. Every automatic action is recorded with its reason.

The dashboard

Opens on any phone, tablet or wall-mounted screen. Live status, controls, forecasts, and Ask A.D.A., an assistant that explains what happened in English or Pidgin.



Power panel: source, live voltage against the safe band, battery, solar, prepaid units, fuel and a 24-hour source history. Sample data.



Forecast panel: battery runtime, when light usually goes, and days of units and diesel left. Sample data.

FOR FACILITY MANAGERS

What you get, property by property

FEATURE	WHAT IT MEANS FOR YOUR TEAM
Live power source	See instantly whether each building is on NEPA, generator, inverter or has no power, with live voltage. Every changeover is timestamped.
Generator hours & diesel	Exact run-hours per generator per day, with estimated diesel use and cost. A clean record for reconciliation with vendors and residents.
Borehole automation	The pump starts when the overhead tank is low and power is available, and stops at 95%. No overflow, no dry tanks, no one assigned to the switch.
Load shedding on inverter	Heavy loads switch off automatically when a building drops to inverter, so batteries last through the night and last more years.
Voltage protection	Sensitive equipment is cut during spikes above the safe band, and the event is logged with the reading.
Lights & security	Compound and security lights switch on during outages or on a schedule. Gate and lock status on the same screen.
Forecasts	Battery runtime at the current load, when outages usually start, and when diesel or prepaid units will run out.
Clear event log	Every automatic action in plain language with its reason, e.g. "pump off: tank at 95%" or "TV off: spike 304 V".
Honest data	If a building's controller stops reporting, the dashboard says the data isn't live and refuses to switch anything, instead of showing stale numbers.
Access per person	Owners, managers and staff each get the dashboards they need, with off, view or edit rights.

SAMPLE WEEKLY REPORT · ONE BUILDING

BLOCK C · WEEK 45 — SAMPLE FIGURES, FOR ILLUSTRATION

GRID (NEPA)	GENERATOR	INVERTER	PUMP CYCLES
71 h	**19 h**	**78 h**	**11**
42% of the week	≈ 68 L diesel	battery never below 30%	0 overflows · 0 dry tank

6 of the 19 generator hours ran while the inverter still had more than 60% charge. Starting the gen later on those evenings would have saved about 21 L of diesel. 2 voltage spikes (296 V, 304 V) were caught; sensitive loads were cut both times.

HOW IT WORKS

From the distribution board to your phone

Power sources
NEPA, generator, inverter, solar

→

A.D.A. controller
Senses the source and voltage; relays switch the pump and heavy loads

→

A.D.A. hub
Rules, event log and forecasts, on site, offline-capable

→

Dashboard
Phone, tablet or wall screen; Ask A.D.A.

SAFETY, BY DESIGN

- **Fails safe.** If the controller can't reach the hub for 10 seconds, every relay drops to its safe state: pump off, heavy loads off.
- **No silent queueing.** If readings are more than 5 seconds old, the dashboard says so and refuses switch commands.
- **Qualified wiring only.** Any work on mains wiring is carried out by a qualified electrician.
- **Rules stay in charge.** A manual switch that breaks a rule (e.g. AC on while on inverter) is reversed, and the reason is shown.
- **Data stays on site.** The hub keeps the log locally. Nothing leaves the building unless you turn on remote access or the online assistant.

ASK A.D.A. · ENGLISH AND PIDGIN

How long will the inverter last?

At 1,700 W the battery (78%) lasts about 2 h 18 m. With the TV and bedroom AC off, about 8 h. A.D.A. does this automatically when it switches to inverter.

Why pump off?

E off for 14:58 because tank don full reach 95%. Pump go start again by itself for 20%, as long as NEPA or gen dey.

Why was the TV switched off at 17:40?

The voltage jumped to 304 V. Anything above 260 V can damage electronics, so A.D.A. cut the TV immediately.

Answers come from the building's own live data. Forecasts and basic answers work offline; the full assistant uses Claude when there is internet.

WHAT GETS INSTALLED

ITEM	PURPOSE
Controller unit	ESP32-based board in an enclosure near the distribution board, with relay outputs for the pump and selected loads.
Sensors	Mains voltage and source detection, and an ultrasonic level sensor on the overhead tank.
Hub	A small always-on device (or an existing phone/laptop during the pilot) on the building's Wi-Fi.
Dashboard	Nothing to install: it opens in a browser on any phone, tablet or wall screen.

THE PILOT

One building, about six weeks

WEEK 0
Pick the building
Together we choose one block or building where power and water cause the most trouble, and agree what success looks like.

WEEK 1
Install
Controller, tank sensor and hub are fitted. Any mains connection is done by a qualified electrician. Dashboard access for your team.

WEEKS 2–5
Run & observe
A.D.A. logs every changeover, generator hour and pump cycle, and handles the routine switching. Weekly check-ins.

WEEK 6
Review
We go through the logs, diesel and water savings, and gaps, and decide together on next steps.

What we need from you
Access to one building and its distribution board · the hardware parts, from a list we provide (funding agreed up front) · a contact on site · Wi-Fi or a hotspot · honest feedback at the weekly check-ins.

What you get
A complete record of the building's power and water for six weeks · automated pump and load control · weekly summaries · a say in how A.D.A. is shaped for facility managers.

WHERE WE ARE

- ■ Hub with automation rules and event log
- ■ ESP32 controller reporting live to the hub
- ■ Fail-safe and staleness behaviour
- ■ Dashboard for phone, tablet and wall screen
- ■ Forecasts and Ask A.D.A. (English & Pidgin)
- ☐ Tank-level and mains-voltage sensors
- ☐ Relay switching of real loads, installed by an electrician
- ☐ Multi-building view for facility managers
- ☐ Weekly report by email or WhatsApp
- ☐ First pilot building

■ Working today ☐ Next, shaped by the pilot

COMMON QUESTIONS

Does it need internet?
No. The rules, log and forecasts run on site. Internet adds remote access and the full assistant.

What if the controller fails?
Relays fall back to safe states, and the dashboard shows the building as not live.

Does it replace the inverter or changeover?
No. It works alongside your existing gen, inverter and changeover, and manages loads around them.

Who sees the data?
Only people you give access to, with off, view or edit rights per dashboard.

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